

JINCHAO SONG

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EDUCATION

2020 Ph.D. Department of Geosciences and Natural Resource Management, University of Copenhagen
2015 M.Sc. Chinese Academy of Sciences
2012 B.A.Sc. Ludong University, China

ACADEMIC EXPERIENCES

2024 – Current Postdoctoral research fellow, University of Michigan
2021 – 2024 Postdoc, University of Southern Denmark
2020 – 2021 Postdoctoral Researcher, Institute of GIS and Remote Sensing, Peking University.
2015 – 2016 Research Assistant, Institute of Urban Environment, Chinese Academy of Sciences.

SELECTED RESEARCH HIGHLIGHT

Nature Cities (2026) | First author

Weighing 600 million buildings reveals global urban material inequality and efficiency paths.

Global mapping of building material stocks across more than 600 million buildings, revealing how urban form shapes material inequality and resource efficiency. The study was accompanied by a [Nature Cities Research Briefing](#) featuring editorial commentary by Senior Editor Wei Chen and expert opinion by Ruidong Chang (Adelaide University).

PEER-REVIEWED JOURNAL ARTICLES

- Song, J.,** Sun, K., Goldstein, B., Tong, X., Wang, L., Helbich, M., Liu, Q., Dai, M., Li, X., & Liu, G. (2026). Weighing 600 million buildings reveals global urban material inequality and efficiency paths. *Nature Cities*. <https://doi.org/10.1038/s44284-026-00510-3>
- Liu, Q., Ouyang, X., Zhu, W., Sun, K., **Song, J.,** Li, X., Li, Y., Chen, W., & Liu, G. (2025). Spatiotemporal Characteristics of Global Building Material Intensity Revealed for Circular and Low-Carbon Construction. *Environmental Science & Technology*.
- Sun, K., Li, Q., Dai, M., **Song J.,** Liu G. *et al.* A global dataset for steel aluminum and cement in-use stocks at 500 m gridded level 2000-2019. *Sci Data* **12**, 1560 (2025).
- Sun K, Li Q, Liu Q, **Song J,** Dai M, Qian X, Gummidi SR, Yu B, Creutzig F, Liu G. Urban Fabric Decoded: High-Precision Building Material Identification via Deep Learning and Remote Sensing. *Environmental Science and Ecotechnology*. 2025 Feb 3:100538.
- Song, J.,** Chen, W., Zhang, J., Huang, K., Hou, B., & Prishchepov, A. V. (2020). Effects of building density on land surface temperature in China: Spatial patterns and determinants. *Landscape and Urban Planning*, 198, 103794.
- Song, J.,** Zhao, C., Lin, T., Li, X., & Prishchepov, A. V. (2019). Spatio-temporal patterns of traffic-related air pollutant emissions in different urban functional zones estimated by real-time video and deep learning technique. *Journal of Cleaner Production*, 117881.
- Song, J.,** Zhao, C., Zhong, S., Nielsen, T. A. S., & Prishchepov, A. V. (2019). Mapping spatio-temporal patterns and detecting the factors of traffic congestion with multi-source data fusion and mining techniques. *Computers, Environment and Urban Systems*, 77, 101364.
- Song, J.,** Tong, X., Wang, L., Zhao, C., & Prishchepov, A. V. (2019). Monitoring finer-scale population density in

- urban functional zones: A remote sensing data fusion approach. **Landscape and Urban Planning**, 103580.
- Tang, C., Yang, C., Cai, R. S., Ye, H., Duan, L., Zhang, Z., Shi, Z., Lin, K., **Song, J.**, Huang, X., Zhang, H., Yang, J., & Cai, P. (2019). Analysis of the relationship between electromagnetic radiation characteristics and urban functions in highly populated urban areas. **Science of the Total Environment**, 654, 535-540.
- Song, J.**, Lin, T., Li, X., & Prishchepov, A. (2018). Mapping Urban Functional Zones by Integrating Very High Spatial Resolution Remote Sensing Imagery and Points of Interest: A Case Study of Xiamen, China. **Remote Sensing**, 10(11), 1737.
- T. Lin, X. Liu, **J. Song** et al., "Urban waterlogging risk assessment based on internet open data: A case study in China," **Habitat Int.**, vol. 71, pp. 88–96, 2018.
- H. Ye, Q. Ren, L. Shi, **J. Song** et al., "The role of climate, construction quality, microclimate, and socio-economic conditions on carbon emissions from office buildings in China," **J. Clean. Prod.**, vol. 171, 2018.
- X. Li, **J. Song**, T. Lin, J. Dixon, G. Zhang, and H. Ye, "Urbanization and health in China, thinking at the national, local and individual levels," **Environ. Heal. A Glob. Access Sci. Source**, vol. 15, 2016.
- Song, J.**, C., Li, X. H., Lin, T., Zhang, G. Q., Ye, H., He, X. Y., & Ge, R. B. (2015). A method of extracting urban built-up area based on DMSP/OLS nighttime data and Google Earth. *Journal of Geo-Information Science*, 17(6), 750-756.

RESEARCH BRIEFINGS

- Liu, G., & **Song, J.** (2026). Global characterization of building-level weight reveals urban material inequality. *Nature Cities*. Research Briefing. <https://doi.org/10.1038/s44284-026-00509-w>.

TEACHING AND MENTORING

- Remote Sensing** (EAS 541.001), University of Michigan. Sensing the city: Remote Sensing and Geo-Big Data in Urban Environmental Studies. Delivered one guest lecture as part of the course.
- Sustainable Urban Systems** (EAS 501.074), University of Michigan (Winter 2026). One guest lecture
- GIS for Engineering Sustainability**, University of Southern Denmark (F22) (February 2022 - June 2022 and 2023). Lecturer (master's, 5 ECTS, English): course planning, lectures, practicals and project supervision.
- Remote sensing in land use (March 2018 and March 2017)**. Geospatial monitoring and modelling using TerrSet.
- ArcGIS (October 2015)** Responsibility: Lecture on geospatial statistics based on ArcGIS software
- Master's research mentoring (graduates): Kunyu Yang** (University of Michigan): technical and manuscript writing guidance throughout the project; corresponding author of her manuscript. **Mias Sommer Schjonberg** (University of Southern Denmark): day-to-day research guidance.

MEDIA AND RESEARCH HIGHLIGHTS

- 2026 **University of Michigan SEAS** (17 September). ["Weighing 600 million buildings to help construct a more efficient future"](#).
- 2026 **Anthropocene** (15 September). ["Why less steel and concrete isn't the same as more sustainable"](#).
- 2026 **Michigan News** (14 September). [Weighing 600 million buildings to help construct a more efficient future](#).
- 2026 **EurekAlert!** (11 September). [Weighing 600 million buildings to help construct a more efficient future](#).
- 2026 **Tech Xplore** (11 September). <https://techxplore.com/news/2026-09-million-efficient-future.html>
- 2026 **Peking University** (10 September). [Research news on global building material stocks and sustainable urban transitions \(in Chinese\)](#).
- 2026 **The Sun**. "Weight of the world". (Print edition, 2026).

MANUSCRIPTS UNDER REVIEW

- 2026 Goldstein B., **Song J.**, Gummidi B., Elliot T. Spatial tipping points in the metabolism of cities. **Nature Cities** (Under Review)

- 2026 **Song J.**, Du S., Du D., Yang K., Goldstein B., Multi-scale analysis of seasonal thermal dynamics and landscape thresholds across Chinese cities. **Building and Environment** (Under Review)
- 2026 Yang K., **Song J.**, Goldstein B. Mapping resort-driven land transformation and forest carbon loss in the Yucatán Peninsula, Mexico. **Journal of Environmental Management** (Under Review)
- 2026 **Song J.**, Helbich M., Du D., Hou B., Liu J., & Goldstein B. Monitoring urban traffic emissions using high-resolution satellite imagery and near-real-time congestion data. **Sustainable Cities and Society** (Under Review)

CONFERENCE PRESENTATIONS

- 2025 AAG Annual Meeting. Weighing the world's over 600 million buildings reveals stark global disparities and impending material demands in the Global South. **Jinchao Song***
- 2018 International Geographical Union, IGU thematic conference, Moscow. Impact of Functional Zones on Traffic Congestion Using Real-Time Traffic Data. **Jinchao Song***
- 2018 International Geographical Union, IGU thematic conference, Moscow. Modelling Urban Population Distribution Using Multi-Source Remote Sensing Imagery and Location-Based Service Networks. **Jinchao Song***

CONFERENCE POSTERS

- 2026 Gordon Research Conference on Industrial Ecology, Maine, USA. Weighing 600 million buildings reveals global urban material inequality and efficiency paths. **Jinchao Song**
- 2017 American Geophysical Union Fall Meeting, New Orleans. Simulating population density using high resolution imagery and location-based service data. **Jinchao Song**

JOURNAL REVIEWER

Remote Sensing of Environment; Cities; Scientific Data

EDITOR ACTIVITIES

- 2024 **Guest-Editor**, Special Issue on Remote Sensing: Remote Sensing and GIS for Monitoring Urbanization and Urban Health, Xinhua Li, Shihong Du, Peng Jia, Franz W. Gatzweiler, **Jinchao Song**
- 2025 **Guest-Editor**, Special Issue on Remote Sensing: Remote Sensing and GIS for Monitoring Urbanization and Urban Health (Second Edition), Xinhua Li, Shihong Du, Peng Jia, Franz W. Gatzweiler, **Jinchao Song**
- 2025 Guest-Editor, Special Issue on Applied Sciences: Intelligent Understanding and Analysis of Urban Big Data, Shouhang Du, **Jinchao Song**, Xiuyuan Zhang

AWARDS

- 2021 Postdoctoral Fellowship, China Postdoctoral Science Foundation (\$11,000 USD)
- 2020 Boya Postdoctoral Fellowship, Peking University, China (88,000 Euro)
- 2015 Scholarship from Institute of Urban Environment, Chinese Academy of Sciences (1,300 Euro)
- 2014 Scholarship from Institute of Urban Environment, Chinese Academy of Sciences (1,300 Euro)

GEOSPATIAL METHODS AND TECHNICAL SKILLS

GIS and geospatial computing: raster and vector geoprocessing; spatial tiling; multi-source geospatial data integration; Python; ArcGIS.

Remote sensing and image analysis: Landsat/Sentinel-2 image filtering, cloud masking and monthly compositing; land-cover classification; urban functional zone mapping; building footprint extraction and height estimation; Google Earth Engine (GEE); ENVI.

Urban climatology and thermal remote sensing: land surface temperature retrieval; surface urban heat island assessment; seasonal thermal hysteresis analysis; surface energy-balance modeling.

Geospatial machine learning: random forest and XGBoost modeling; k-means clustering; partial dependence analysis; deep-learning object detection and counting in satellite imagery and video (YOLO, Mask R-CNN).

Transport geography and emissions modeling: spatiotemporal analysis of traffic congestion; road network and urban functional zone analysis; traffic emissions estimation from vehicle counts, speeds and emission factors.

Urban morphology and material stock modeling: building-scale material stock estimation; urban form analysis; model validation and uncertainty assessment.

Spatial statistics and geovisualization: GeoDetector; spatial statistical analysis; thematic mapping and scientific visualization.